ADDITIVITY OF THE DEGREE MAP FOR DISJOINT UNIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file weil.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the identification through the pullbacks along the coprojections, and the decomposition of a class used to compute the degree.

Editorial note. The label lemma-trace-disjoint-union names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source weil.tex –label lemma-trace-disjoint-union –start 3407 –end 3429 –local-id 28158 returned RESULT=OK (L28158 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Cohomology of a disjoint union as a product. Same chapter, the lemma that the cohomology of a disjoint union is the product of the cohomologies, with the isomorphism given by the pullbacks along the coprojections. Identifies the target of the degree map with the direct sum of the targets of the two degree maps.
- Decomposition of a class along the coprojections. The identity expressing a class on a disjoint union as the sum of the pushforwards of its pullbacks, established in the proof of the additivity lemma. Splits the computation of the degree into the two summands.

Source reference. Original reference 0FHJ: sources/stacks-weil/pinned-original-source.tex, source lines 2932–2932. Prerequisite pointer invoked at proof line(s) 3416; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Assume given (D0), (D1), (D2), and (D3) satisfying (A), (B), and (C). Let X, Y be nonempty smooth projective schemes both equidimensional of dimension d over k. Then $\int_{X \amalg Y} = \int_X + \int_Y$.*

Proof. Denote $i : X \to X \amalg Y$ and $j : Y \to X \amalg Y$ be the coprojections. By Lemma 0FHJ the map $(i^*, j^*) : H^*(X \amalg Y) \to H^*(X) \times H^*(Y)$ is an isomorphism. The statement of the lemma means that under the isomorphism $(i^*, j^*) : H^{2d}(X \amalg$

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$Y)(d) \to H^{2d}(X)(d) \oplus H^{2d}(Y)(d)$ the map $\int_X + \int_Y$ is transformed into $\int_{X \amalg Y}$. This is true because

$$\int_{X \amalg Y} a = \int_{X \amalg Y} i_*(i^*a) + j_*(j^*a) = \int_X i^*a + \int_Y j^*a$$

where the equality $a = i_*(i^*a) + j_*(j^*a)$ was shown in the proof of Lemma 0FHJ. $\square$